

Comparison of individual antioxidants of sperm and seminal plasma in fertile and infertile men.

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OBJECTIVE: To measure individual antioxidants in sperm and seminal plasma from fertile and infertile men to determine if any particular antioxidant is reduced in infertile men. DESIGN: Semen samples were prepared by a discontinuous Percoll gradient to separate sperm and seminal plasma, and the antioxidant concentrations of each were assessed. Samples also were screened for phorbol ester-induced reactive oxygen species (ROS) activity. SETTING: Departments of Obstetrics and Gynaecology, and Clinical Biochemistry, The Queen's University of Belfast, Northern Ireland. PATIENT(S): Fifty-nine male patients attending our infertility center: 18 men whose wives had ongoing pregnancies from IVF with normozoospermic semen profiles, 20 infertile men with normozoospermic and 21 men with asthenozoospermic semen profiles. MAIN OUTCOME MEASURE(S): Ascorbate, urate, sulphydryl groups, tocopherol and carotenoid concentrations were measured in sperm and seminal plasma from fertile and infertile men. RESULT(S): In seminal plasma, ascorbate contributes almost twice as much as urate and thiol levels are about one third of ascorbate. Ascorbate levels in seminal plasma of asthenozoospermic individuals (+ROS) are significantly reduced. In sperm, thiols contributed most and ascorbate only a fraction of the total. CONCLUSION(S): In seminal plasma, ascorbate, urates, and thiols are the major antioxidants present. In contrast, within sperm, this group is the major contributor. In samples exhibiting ROS activity, ascorbate concentrations in the seminal plasma are significantly reduced.